

Product Design and Innovation



@CDE2300 Product Design and Innovation

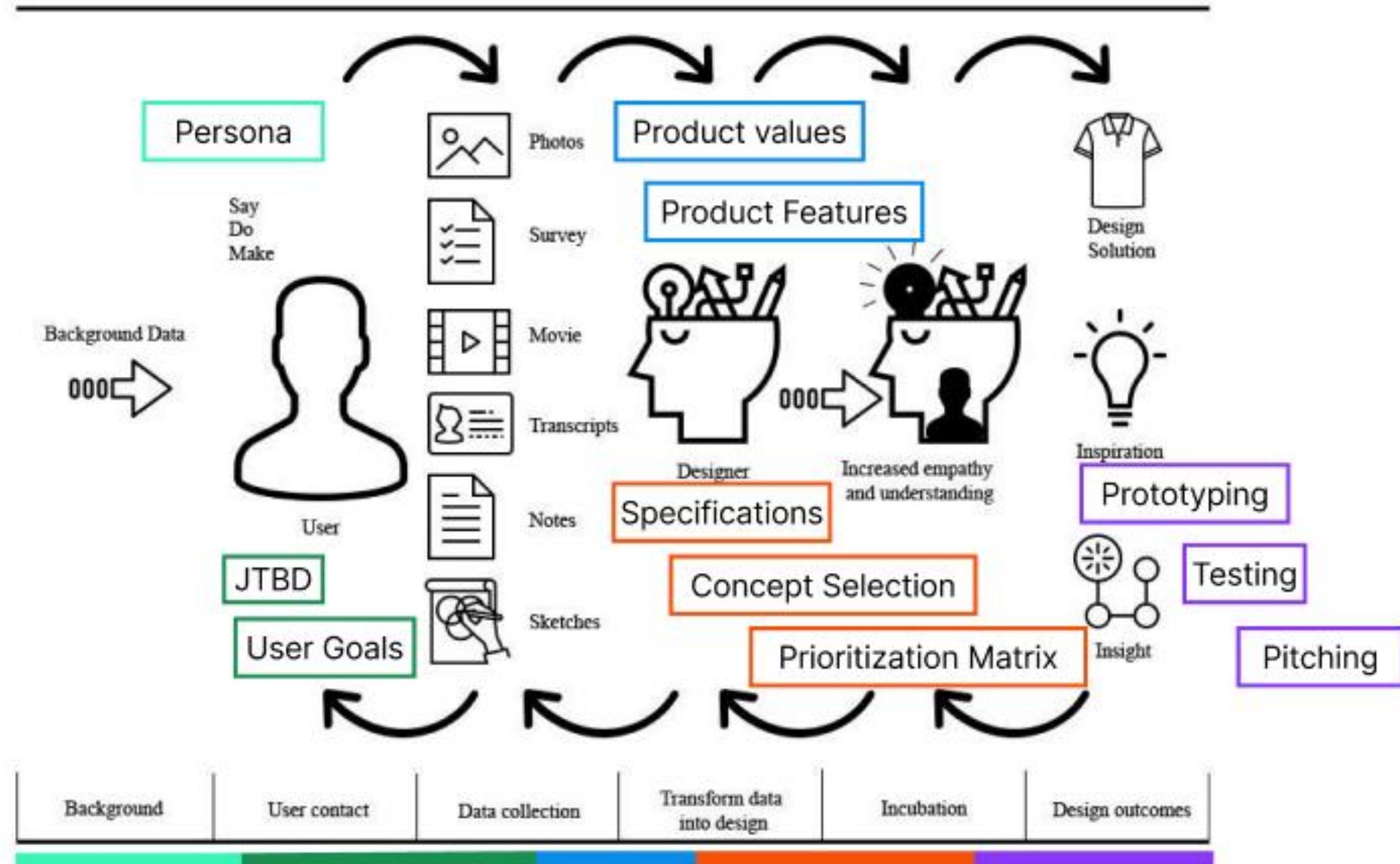


CDE2300 pdi

Course Overview

Week	Titles	Week	Titles
1 (12 Jan)	Innovation-driven economy, Group formation, and Brief intro	7 (2 Mar)	Concept generation, Prioritization map, Concept selection, and Prototyping
2 (19 Jan)	Experience economy, JTBD, Job story and Job statement	8 (9 Mar)	Prototyping and Prototyping Plan
3 (26 Jan)	Empathy Map, Fieldwork, Affinity Clustering, and Persona	9 (16 Mar)	Storytelling
4 (2 Feb)	Ideation methods, Users' goals, Product values, features, and vision, and Hands-on exercise: Crazy 8	10 (23 Mar)	Testing
5 (9 Feb)	Mid Term Review	11 (30 Mar)	Group consultation
6 (16 Feb)	No class!	12 (6 Apr)	Group consultation
23 Feb to 27 Feb	Recess Week (No Class)	13 (13 Apr)	End Term Review (Show and Tell) (2pm to 6pm)

Overview of Design Process



CDE2300 Breakdown of Assessments

Individual	Weightage
Assignments	5%
Assignment 1 (JTBD Statement)	
Assignment 2 (Persona on All the Tea in England)	
Assignment 3 (Prototyping Plan)	
Design Reflection	5%
Peer Reviews	5%
Mid Term Review (with TAs moderation)(2.5%)	
End Term Review (with TAs moderation)(2.5%)	
Group	
A1 Posters	20%
Prototype	28%
Mid Term Review	12%
Final Pitching	25%
CDE2300 Teaching Team (10%)	
Invited Judges (15%)	

EMPATHY MAP

SEE? HEAR?

SAY & DO?

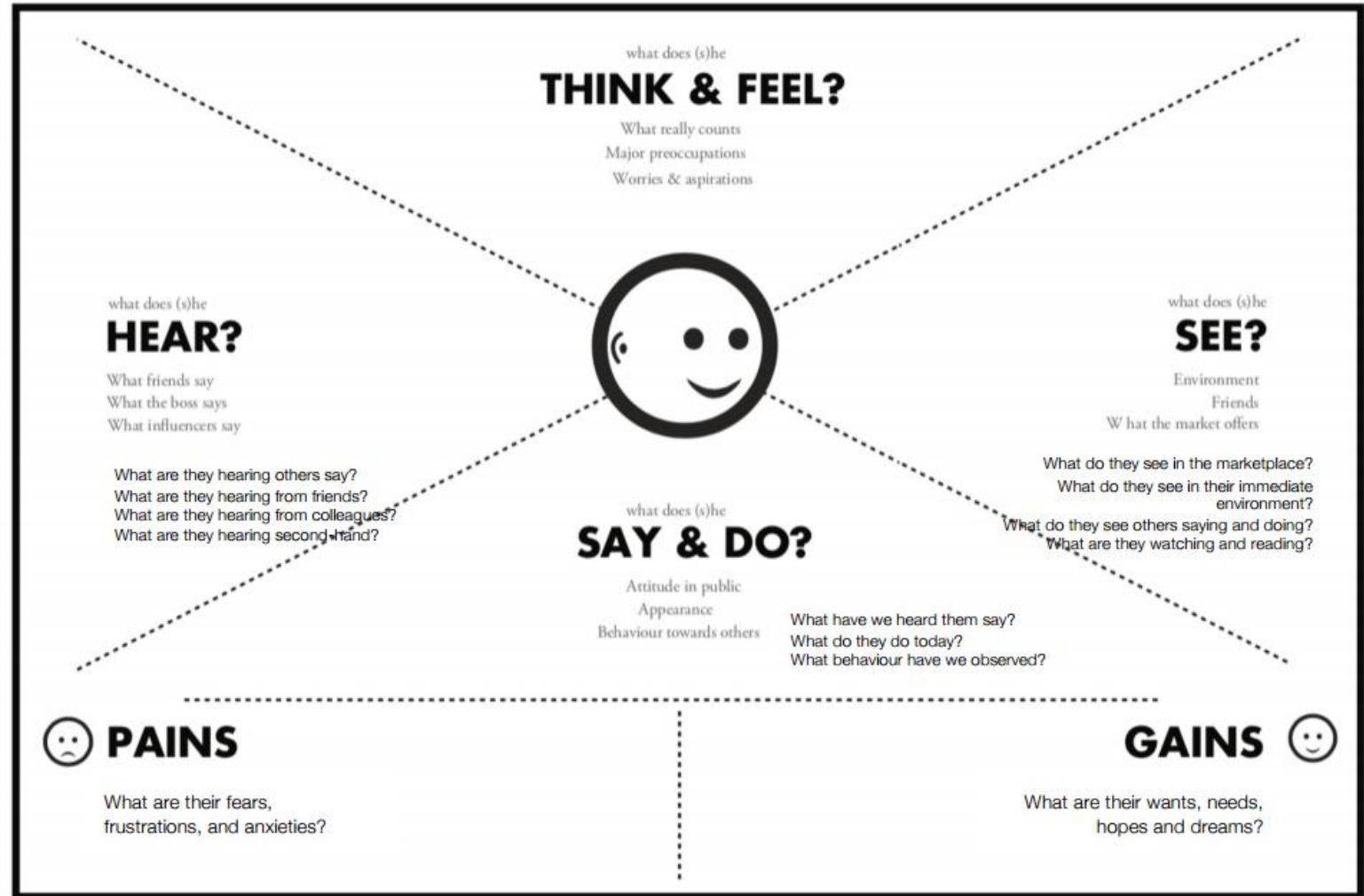
THINK & FEEL?

PAINS

- What are the pain points?
- Challenges?
- Problems or doubts
- Obstacles
- Frustrations

GAINS

- What do users hope to gain?
- What do users want to achieve?
- Goals, Needs
- What is success?
- How success is measured?



Empathy Map Canvas ©2017 David Grey

**Link to softcopy of templates for
Empathy Map, JTBD and POEMS**



https://miro.com/app/board/uXjVJT1GNOW=?share_link_id=398374781407

EMPATHY MAP

SEE? HEAR?

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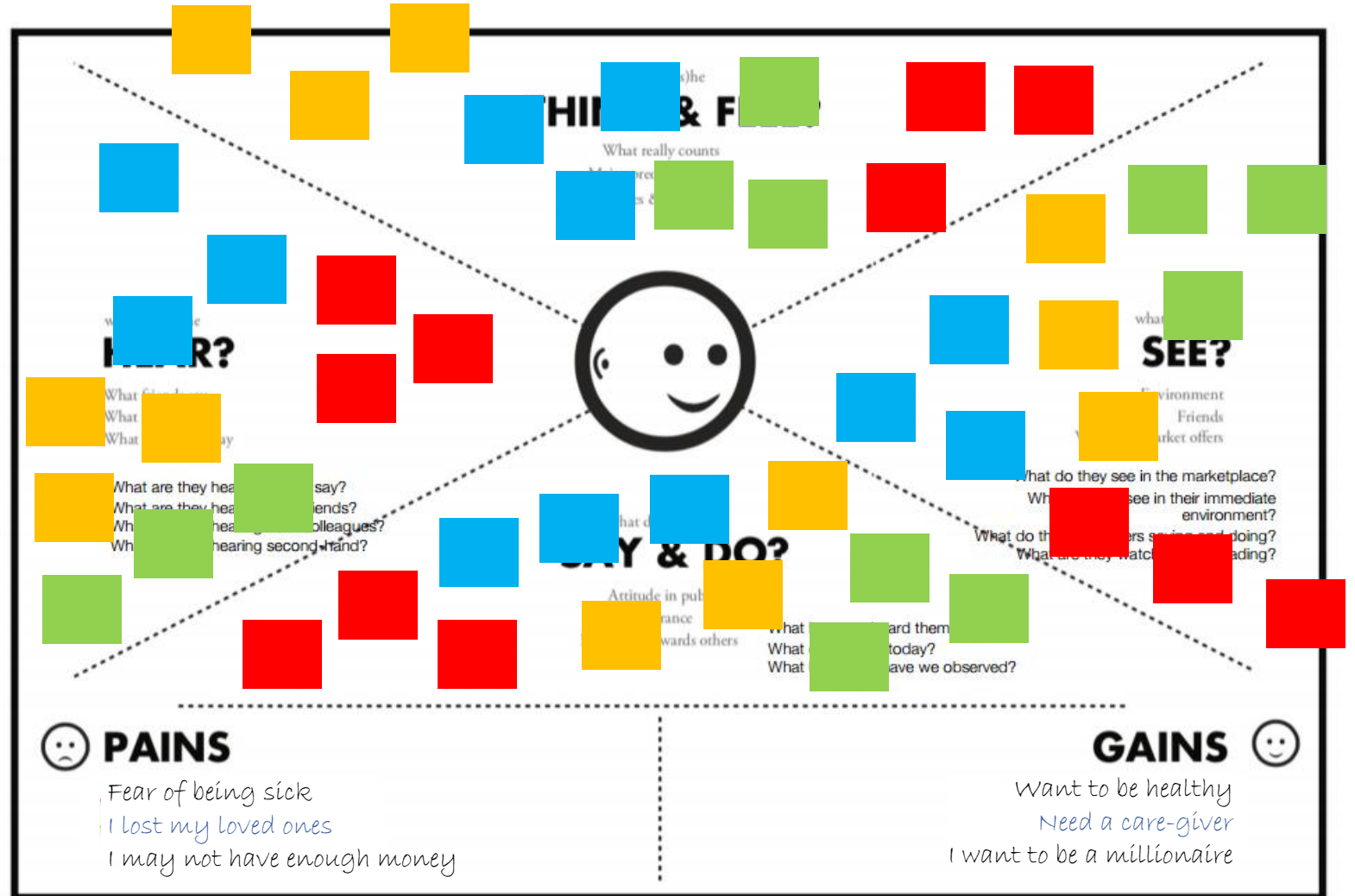
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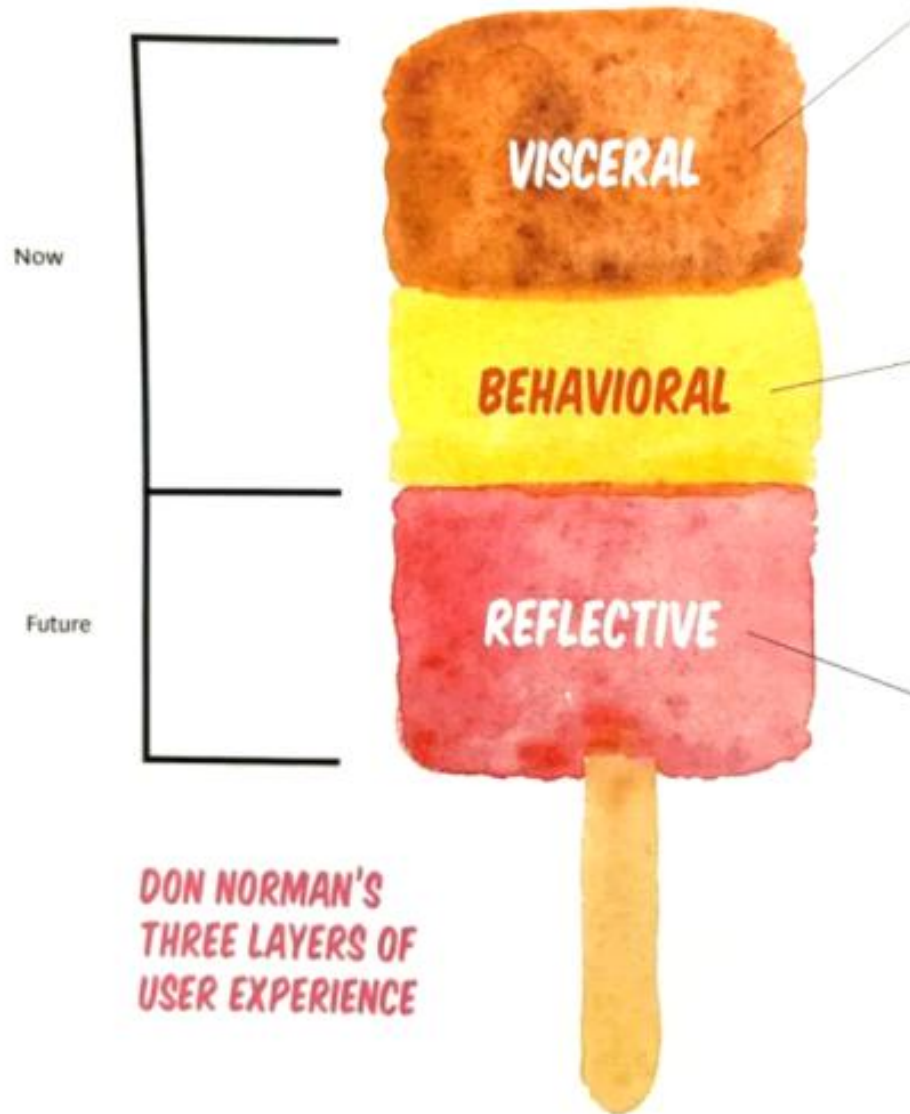
Experience Economy

The emotion wheel (Dr. Robert Plutchik's Wheel of Emotions) is a tool that helps to identify & label user's feelings by categorizing emotions into core, nuanced variations to facilitate better communication.



Experience Economy

This is how differently users respond to the user experience (or a story, a car ride, or anything else) with the passage of time (from the Beginning to the Future).



Beginning

At the visceral level, design stirs an immediate reaction to form, colour, texture and materials used.

Middle

At the behavioural level, design prompts a physical response or action.

End

At the reflective level, design engages memories and associations.

Experience Economy

*Adapted
Maslow's
Hierarchy
[of user
experience]*

Understanding user experience allows the designer to move up the hierarchy (i.e. from Functional to Pleasurable).

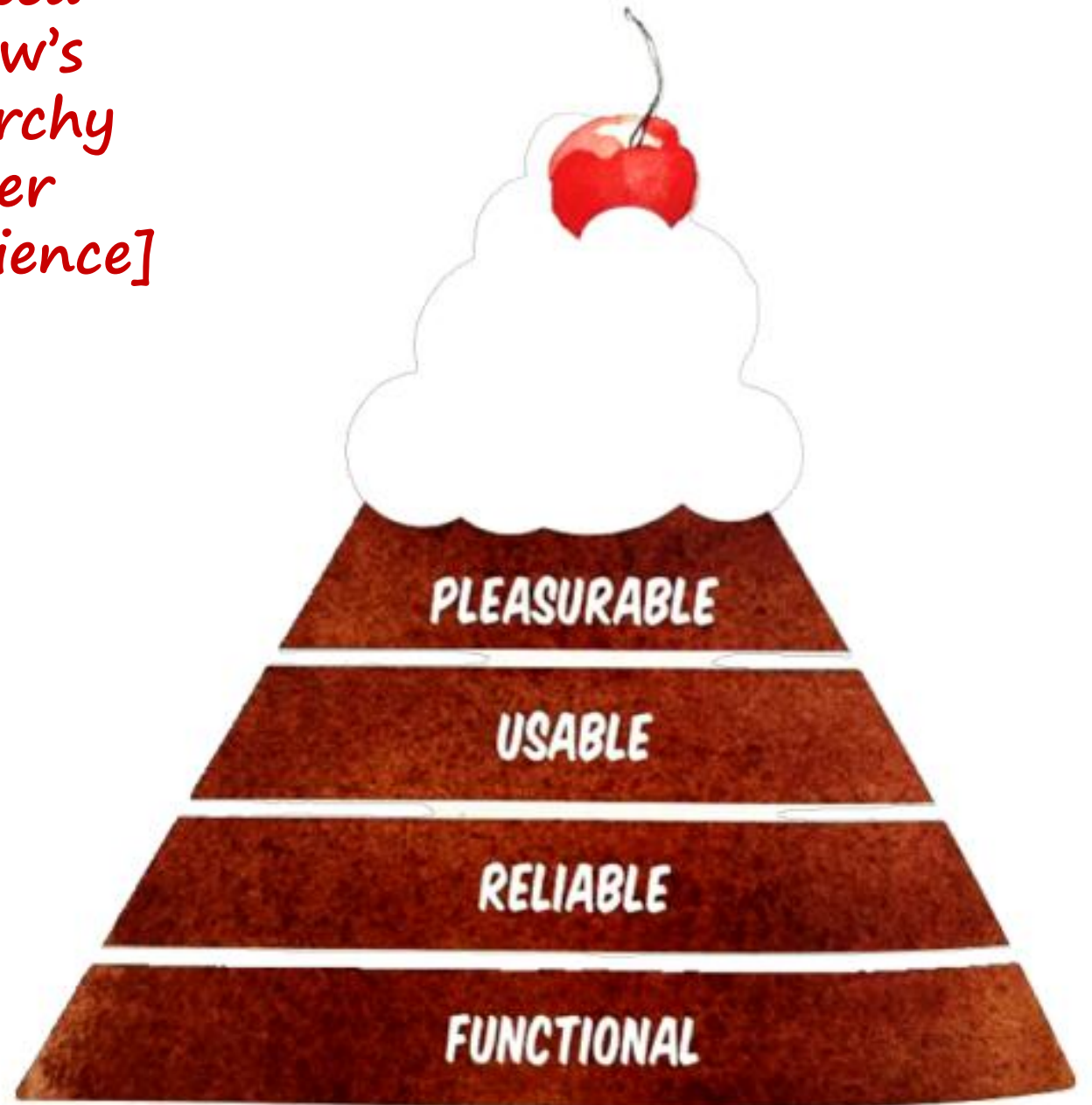
Functional: Does it work? If not, nothing else matters.

Reliable: Can users trust it to work consistently?

Usable: Is it easy and intuitive to use?

Pleasurable: Does it create moments of joy that users remember?

User delight is the cherry on top



Level of Needs Expressed

Explicit needs

can be expressed in words.

Observable needs

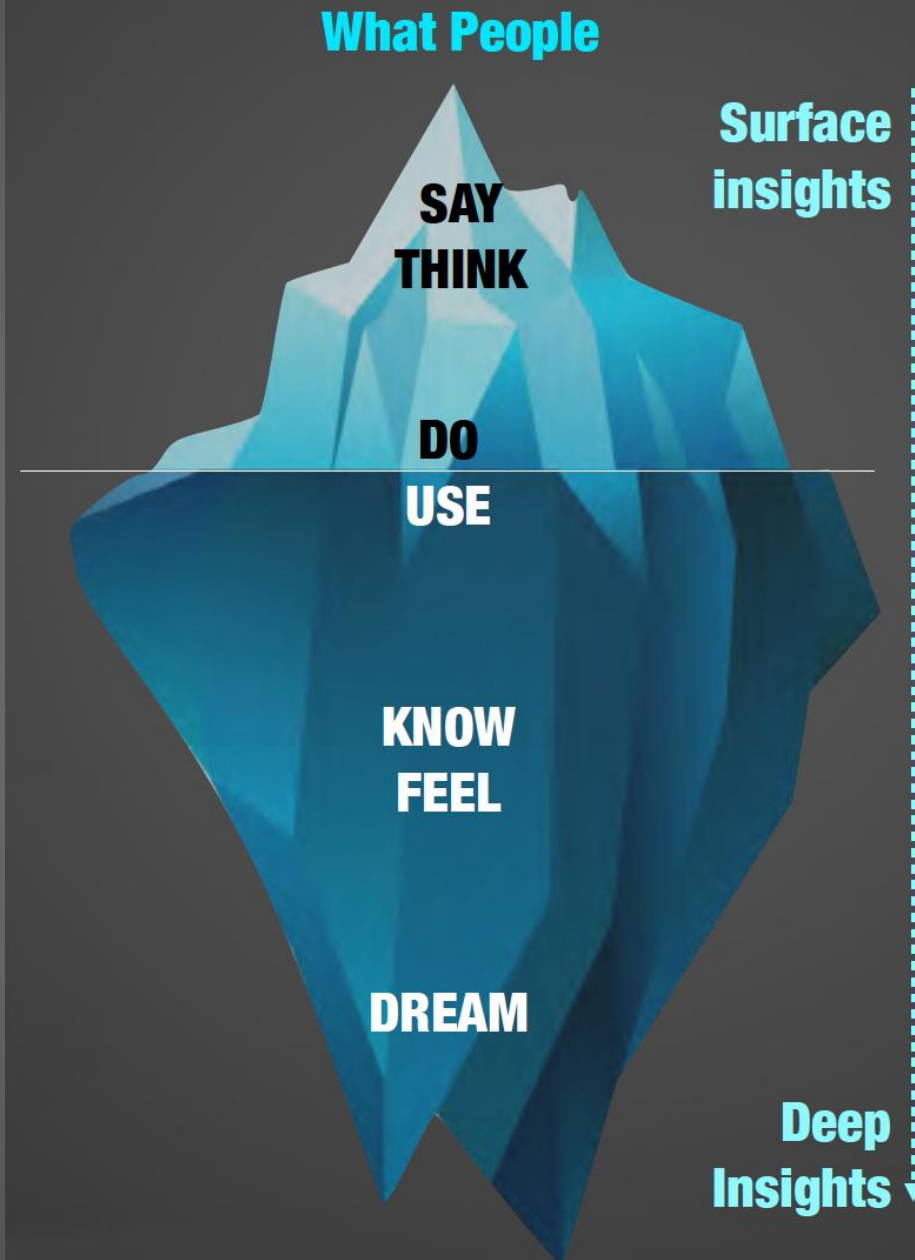
can be uncovered through observation.

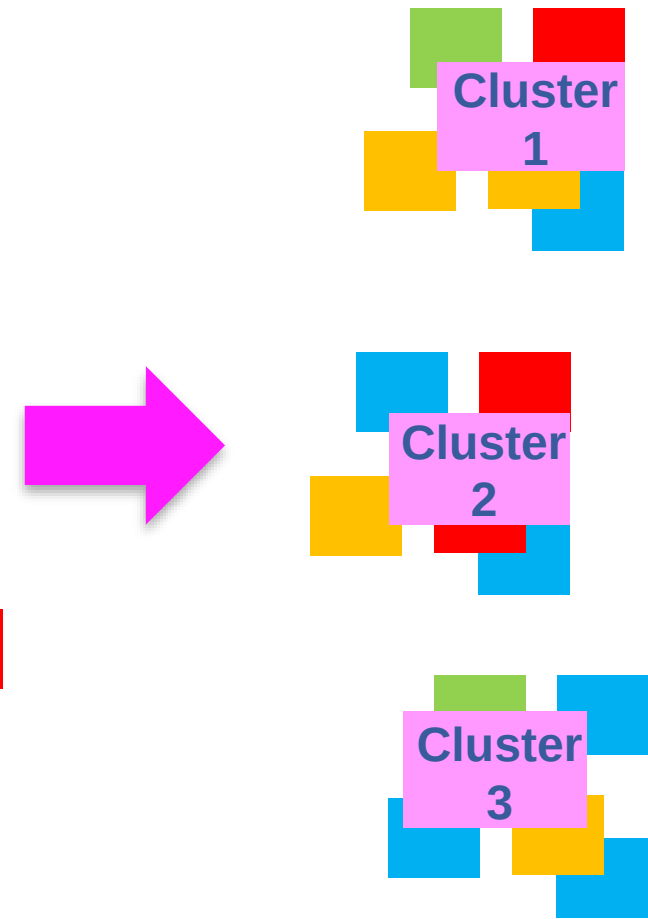
Tacit needs

Are conscious needs that customers are unable to express in words.

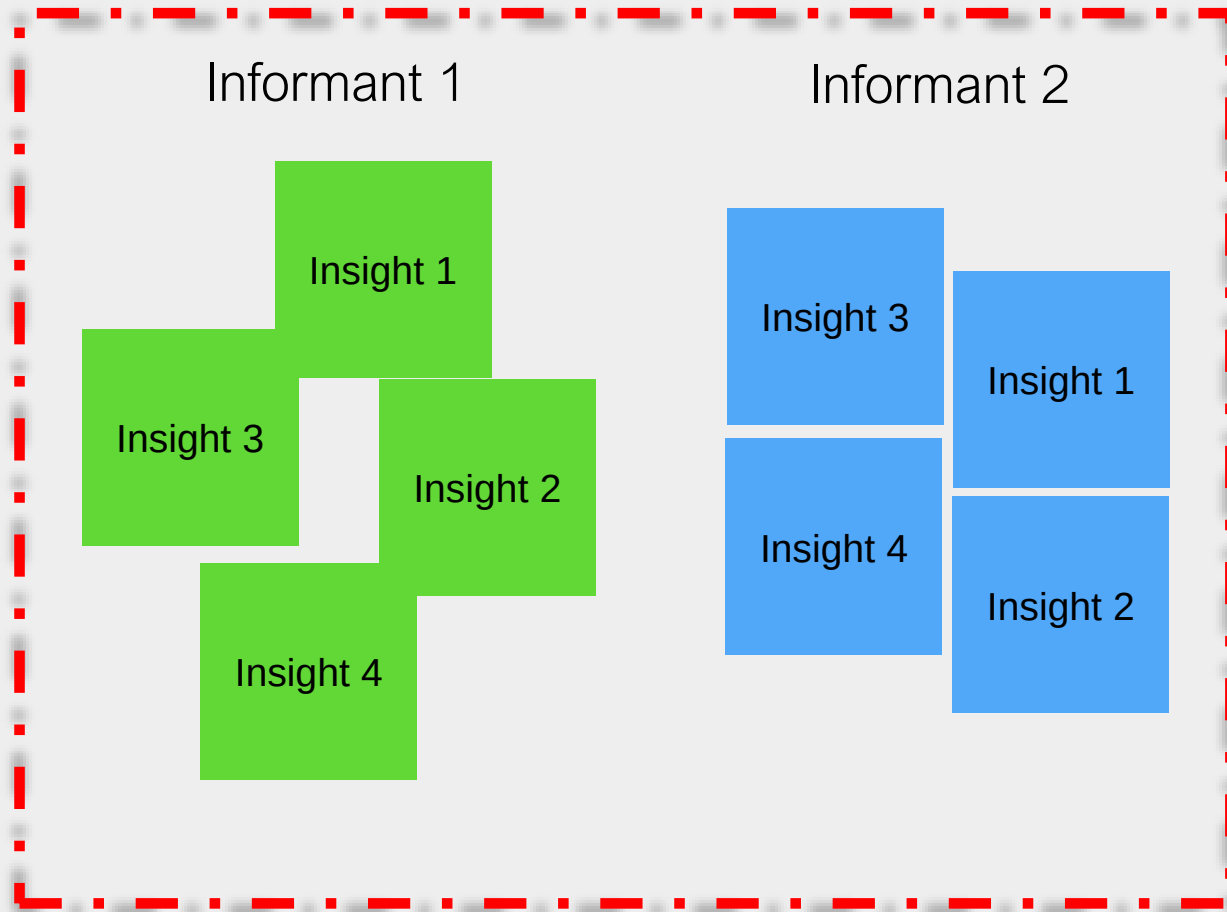
Latent needs

Are subconscious or dormant needs that customer are unable to express in words.



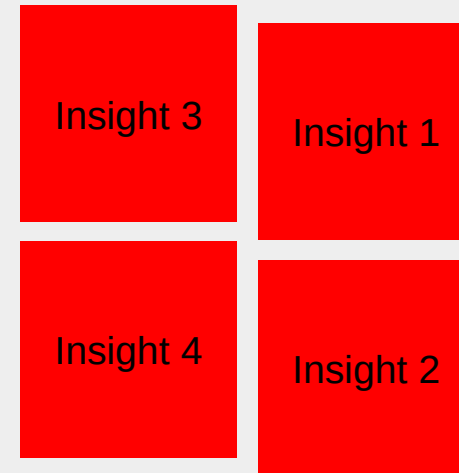


Persona 1



Persona 2

Informant 3



Persona 3

Informant 4



Example of Persona

ALEXIA TEO

TECH-SAVVY PROFESSIONAL

📍 Singapore

ABOUT

Product Manager at a fintech startup

Age: 29

Gender: Female

Location: Singapore (lives in Tiong Bahru, works in Raffles Place)

Occupation: Product Manager

Education: Bachelor's in Computer Engineering

Income: SGD \$90,000 per year

Marital Status: Married, no kids (yet)

📅 TECH HABITS

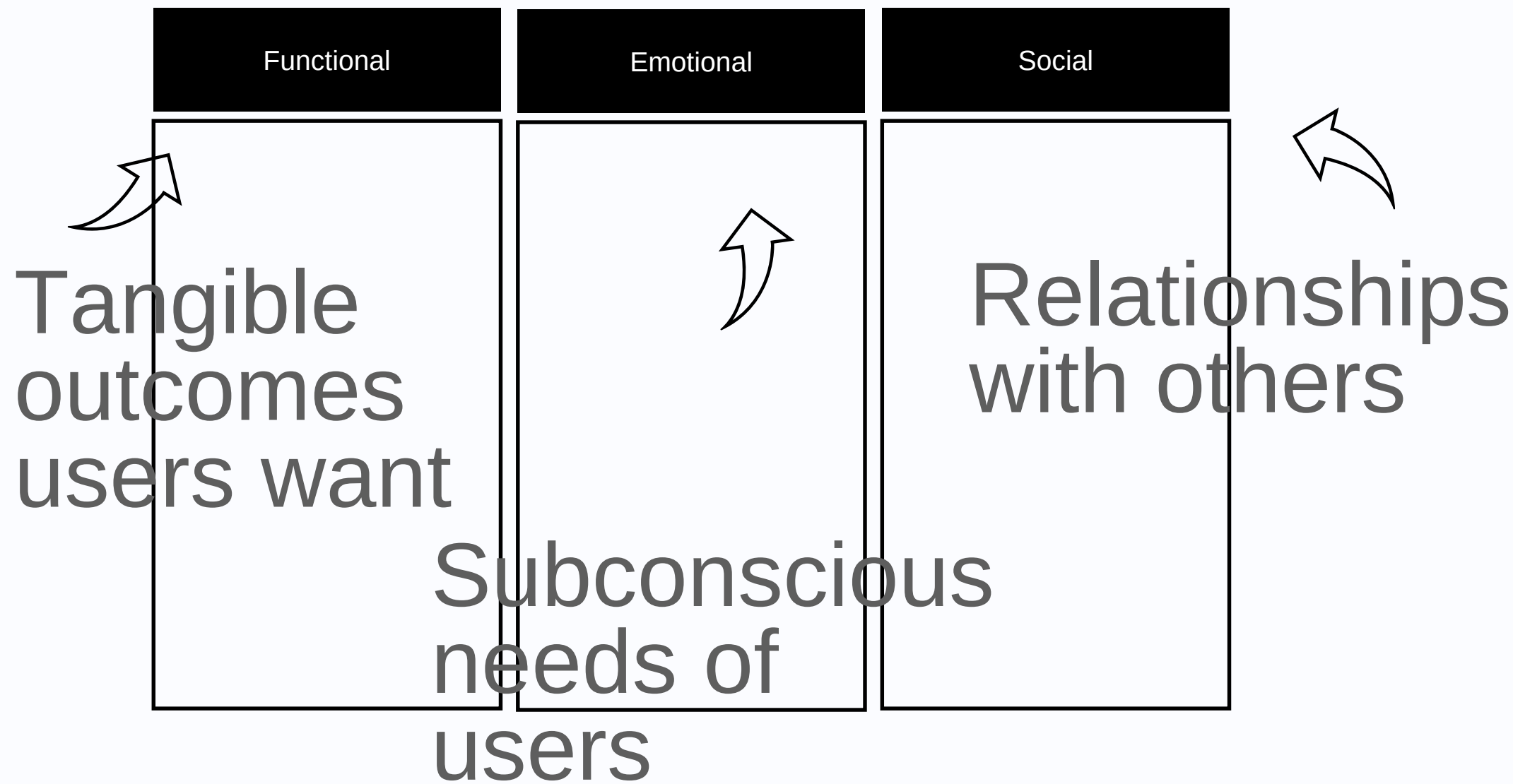
📌 Owns the latest iPhone PRO and MacBook Air M2 for work

LIFESTYLE & ROUTINE

Weekdays: Long work hours (9 am--8 pm), often with late-night calls across time zones



JTBD Template



Job Statement

WHEN I BUT HELP ME SO I



What's the context?



What's the barrier
getting in the way
today?



What is the user
trying to do?



What's the outcome of
achieving this action?



User Goals – Why's behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.

Product Features – How the product delivers

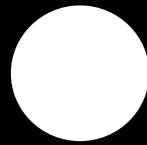
The specific functionalities or capabilities of a product.

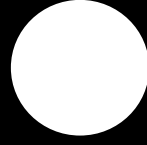
Product Values – What the product delivers

The desired outcomes or objectives that users aim to achieve by using a product or service.

User Goals – Why's behind the JTBD

The desired outcomes or objectives that users aim to achieve by using a product or service.



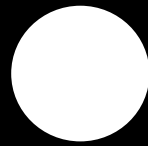


Product Values – What the product delivers

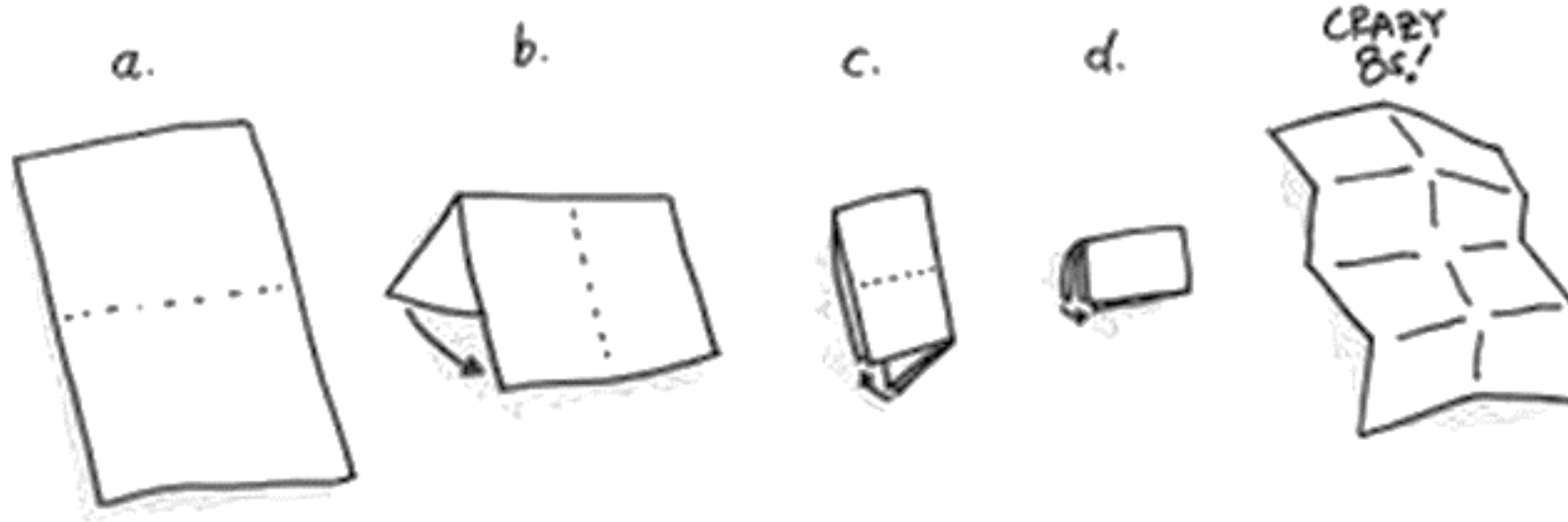
The desired outcomes or objectives that users aim to achieve by using a product or service.

Product Features – How the product delivers

The specific functionalities or capabilities of a product.



Brainstorming



Crazy 8 Ideation
~10 mins

From your list of Oxymorons, pick 1
concept and think of 8 ways you can do it

8mins total, 1mins each segment

Prototyping

Why Prototype?

- Think and Learn,
- Get ideas across,
- Test possibilities,
- Identify weaknesses,
- Fail quickly and cheaply.

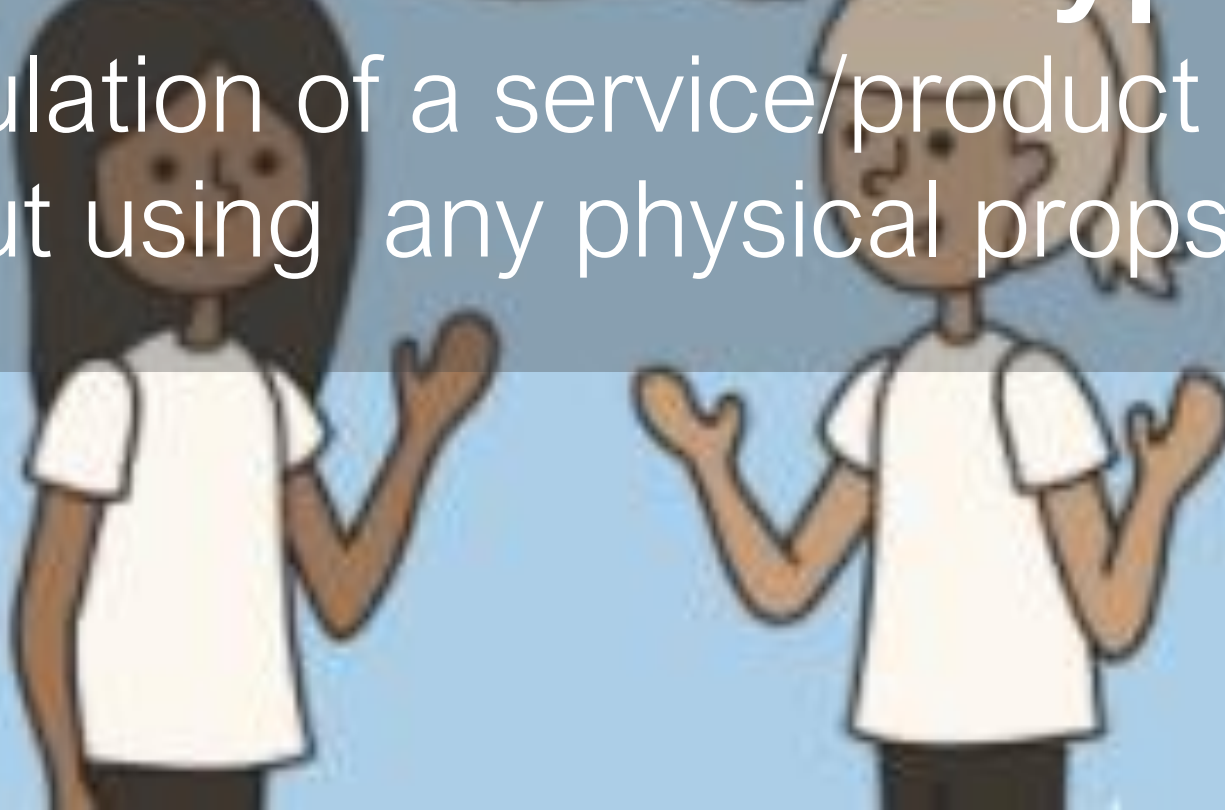
How to Prototype?

- Think about what you are trying to learn by making a prototype,
- **Stay as low-resolution as possible,**
- Remember a prototype doesn't have to be or look like the solution idea.

3 types of prototypes:

1. **NO Resolution Prototyping:**

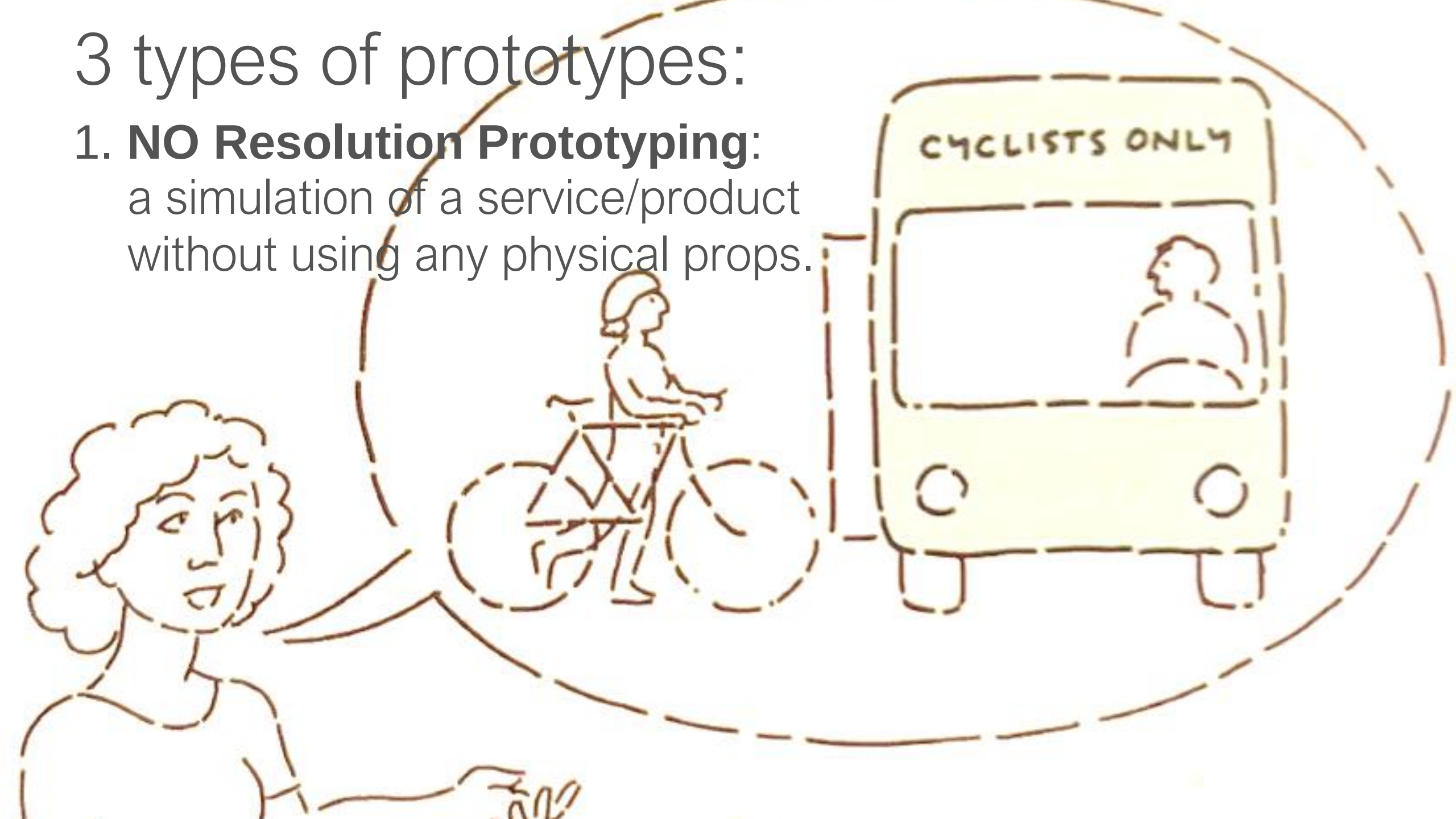
a simulation of a service/product without using any physical props.



3 types of prototypes:

1. **NO Resolution Prototyping:**

a simulation of a service/product without using any physical props.



3 types of prototypes:

2. LOW Resolution Prototyping:

a rapidly built model approximating the look and feel of a product, service or system.

3 types of prototypes:

3. HIGH Resolution Prototyping:

a model containing a high level of realism to define appearance of a product, service or system.

Envisioning Possibilities: Experiments and Prototyping

Inside the Tyler Centre at Starbucks

Link: <https://youtu.be/fLgzDa3gcIs>

Advanced Re-manufacturing and Technology Centre

Industry Sectors

Chemical And Energy

Electronics

Engineering

Food, Nutrition And
Consumer Care

Infocomms

Medical Technology

Pharmaceuticals &
Biologics

ARTC

Advanced Re-manufacturing and Technology Centre by
A*Star



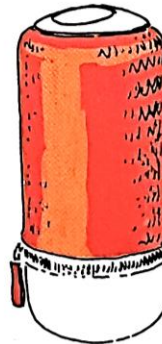
Examples of Prototypes



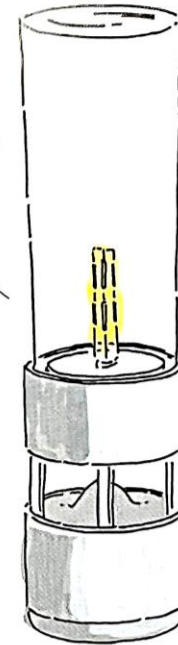
Examples of Prototypes

PRODUCT PERSONALITY PROFILE— TWO WIRELESS SPEAKERS

Zipper allows
users to change
the bright cover;
this product
is moderately
priced.



A warm, dimmable
LED light inside the
glass cylinder resembles
a candle flame; this
product is expensive.



PROFILE #1

Libratone ZIPP Mini
wireless speaker

PROFILE #2

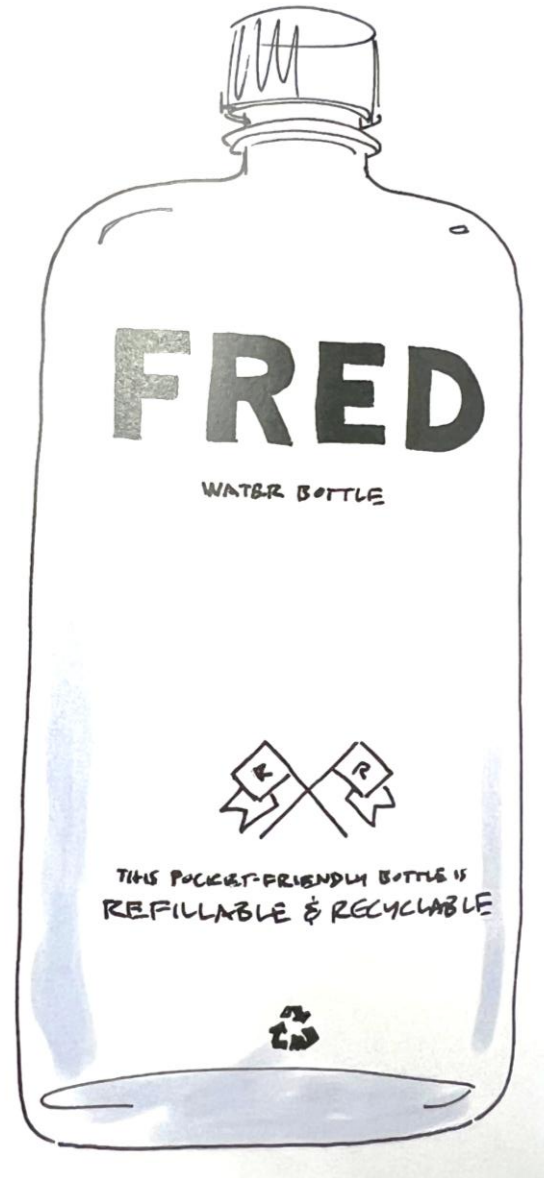
SONY Glass
wireless speaker

TARGET USER

AGE	25
OCCUPATION	programmer
HOUSING	urban apartment
TRANSIT	bicycle
PERSONALITY	dynamic/extrovert
VACATION	hiking
HOME STYLE	IKEA and thrift
CLOTHING	jeans and vintage

AGE	45
OCCUPATION	fashion buyer
HOUSING	3-bedroom townhouse
TRANSIT	BMW
PERSONALITY	organized
VACATION	Euro-camping
HOME STYLE	upscale contemporary
CLOTHING	Club Monaco

Examples of Prototypes



Examples of Prototypes

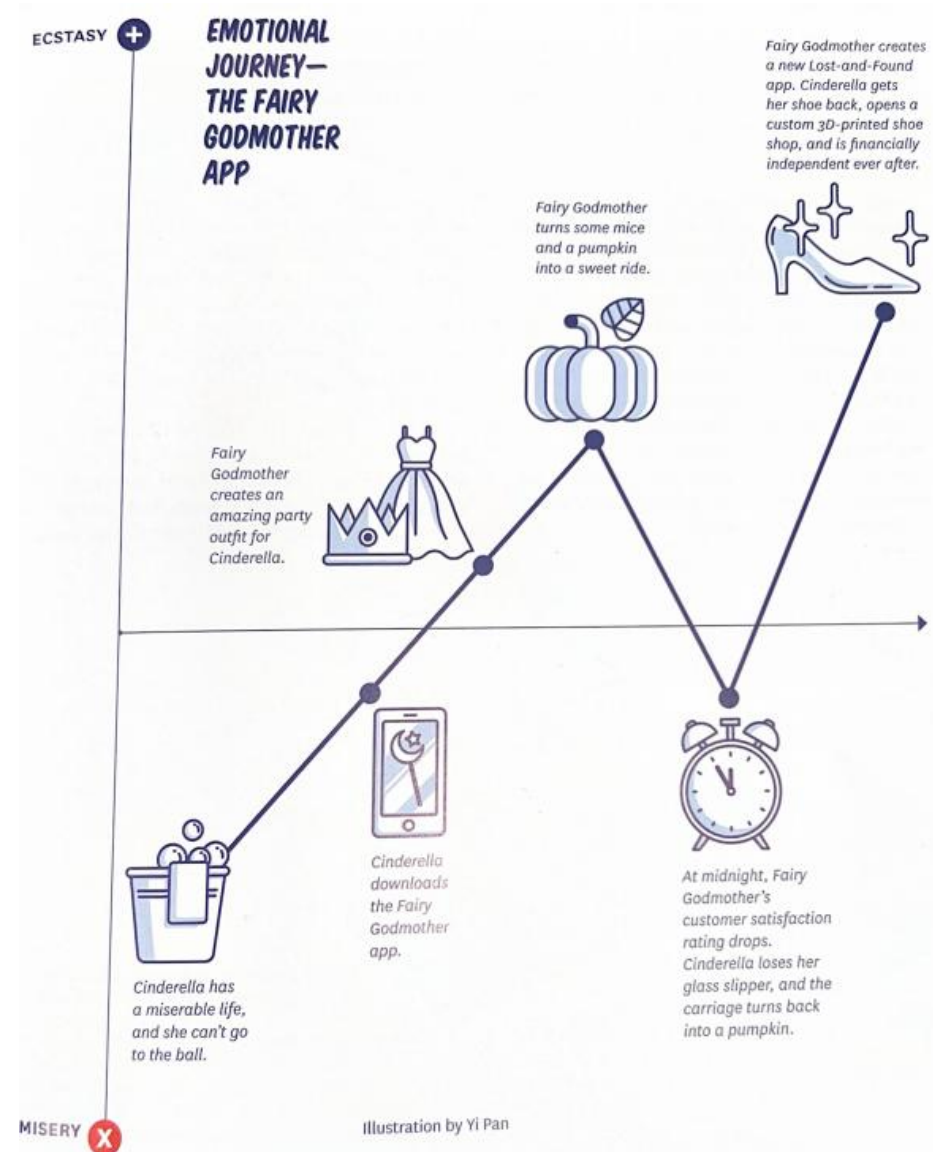
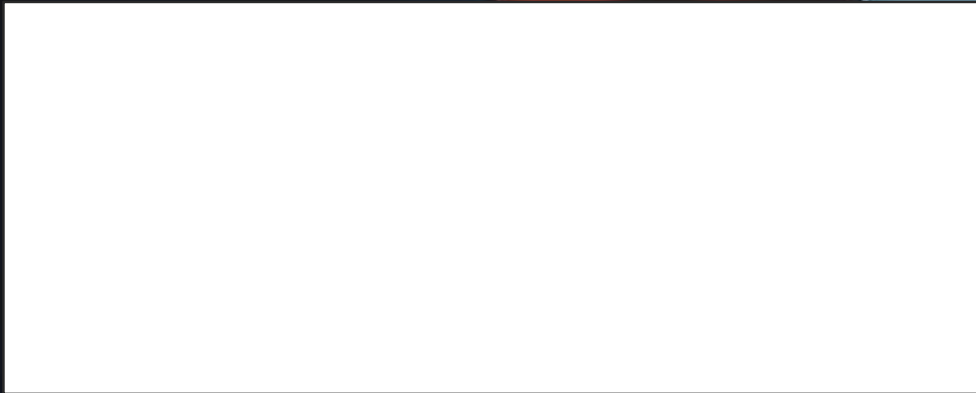


Illustration by Yi Pan

Your prototyping plan

1. Your design problem/design challenge

A large, empty white rectangular box intended for the user to describe their design problem or challenge.

2. Your solution

A large, empty white rectangular box intended for the user to describe their solution.

3. Your prototyping plan for user testing

A large, empty white rectangular box intended for the user to describe their prototyping plan for user testing.

My prototyping plan

1. Your design problem/design challenge

Describe your design problem/design challenge in less than 200 words.

2. Your solution

Describe your solution or sketch it out.

3. Your prototyping plan for user testing

- What is the most important assumption you and your team need to test? (This is the assumption that will cause the entire solution to fail if it is wrong!)
- What are the prototype(s) that you will build to test the solution? (Which features?)
- What are the tools and materials needed?

Describe your design problem/design challenge in less than 200 words.

Describe your solution or sketch it out.

- What is the most important assumption you and your team need to test? (This is the assumption that will cause the entire solution to fail if it is wrong!)
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Individual

Assignment 3: Your prototyping plan

1. Your design problem/design challenge

Describe your design problem/design challenge in less than 200 words.

2. Your solution

Describe your solution or sketch it out.

3. Your prototyping plan for user testing

- What is the most important assumption you and your team need to test? (This is the assumption that will cause the entire solution to fail if it is wrong!)
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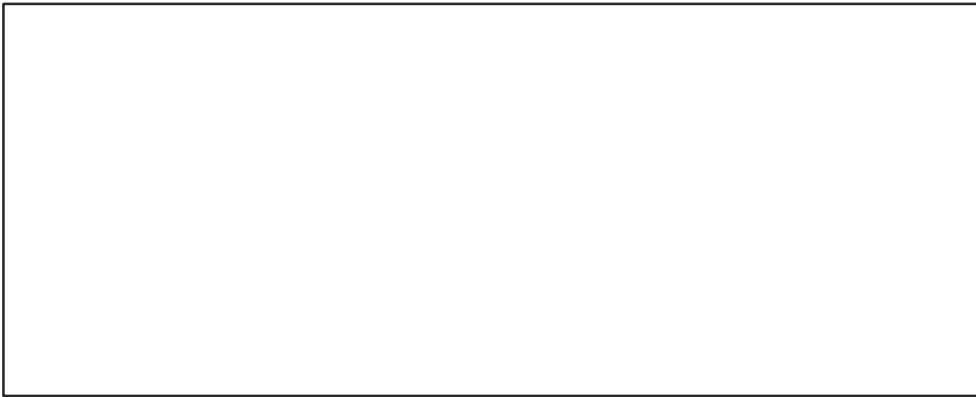
**Link to softcopy of templates for
Empathy Map, JTBD, POEMS,
Prototyping Plan**



https://miro.com/app/board/uXjVJT1GNOW=?share_link_id=398374781407

Your prototyping plan

1. Your design problem/design challenge

A large, empty rectangular box with a thin black border, intended for writing the design problem or challenge.

2. Your solution

A large, empty rectangular box with a thin black border, intended for writing the solution.

3. Your prototyping plan for user testing

A large, empty rectangular box with a thin black border, intended for writing the prototyping plan for user testing.

Assignment 3: Your prototyping plan

4. Additional Information

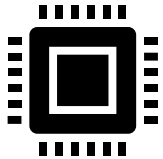
Individual

Assignment 3: Your prototyping plan (Sample)

1. Your design problem/design challenge

HMW design a middleware to allow ferry craft owners to convert their crafts into autonomous ones easily and in low cost.

2. Your solution



This low-cost system-on-chip can be connected to existing diesel powered crafts easily..

3. Your prototyping plan for user testing

- Our solution is cheaper than a new electric craft.
- Scaled-down version of our autonomous craft.
- Quick prototyping tools from our makers' hub.



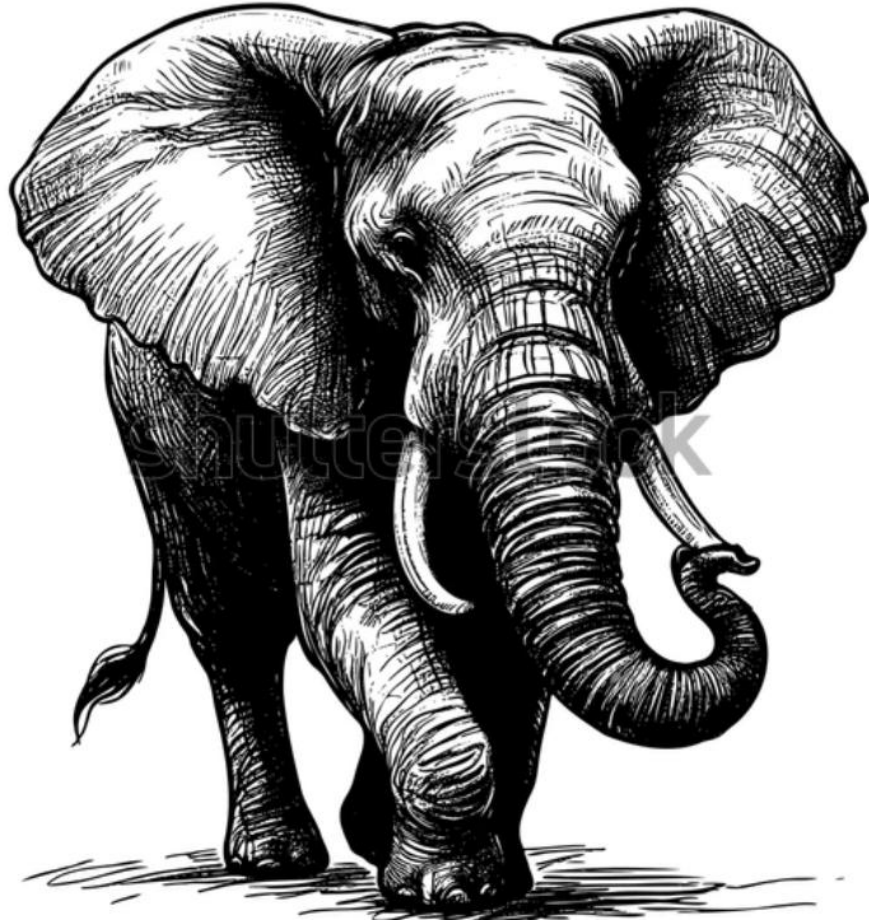
CAD of our prototype

Please name your file using your email address. For example, e1234567.pdf and submit on the respective folder “Assignment 3” on Canvas (Due: 17 March 2026)

Analogy in Design

Metaphor in Design

Big Picture



Analogy in Design

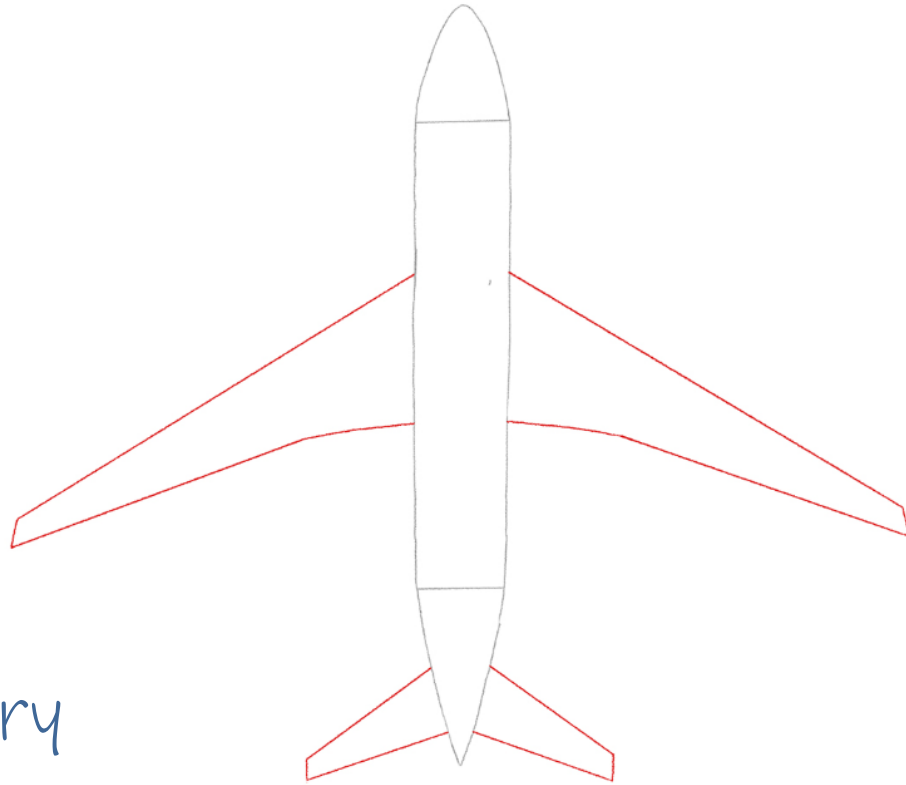


How would an elephant solve the problem?

Examples of Analogy in Design



Biomimicry



Examples of Analogy in Design



User Interface Design

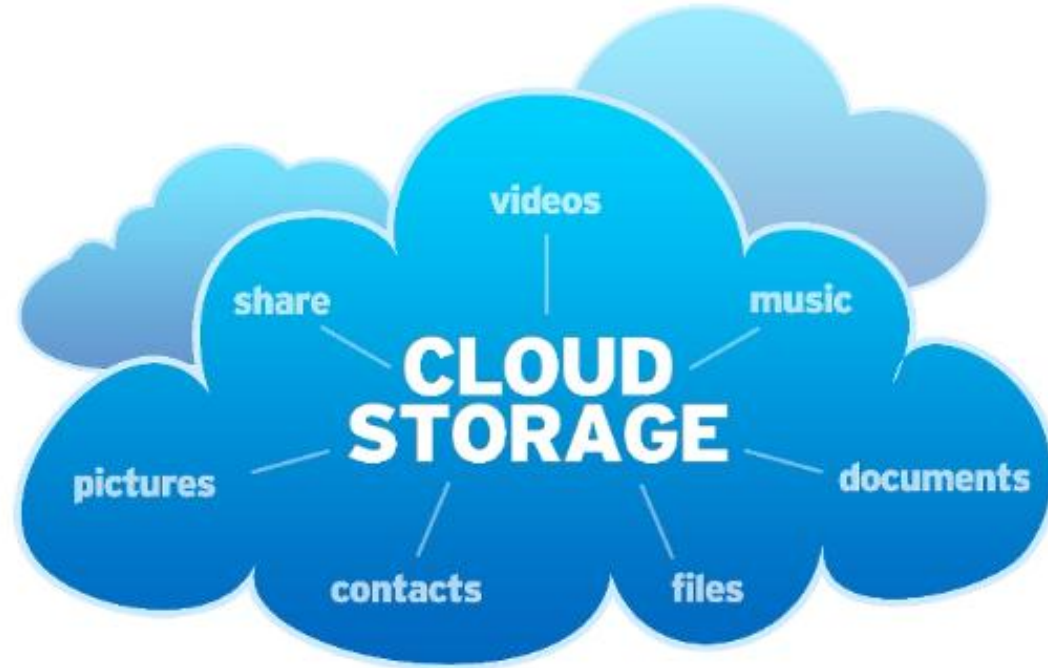
Big Picture

Metaphor in Design

Using familiar concepts to represent more complex or abstract ideas

Big Picture

Metaphor in Design



Using familiar concepts to represent more complex or abstract ideas

Examples of Metaphor in Design



Using familiar concepts to represent more complex or abstract ideas

Examples of Metaphor in Design



Using familiar concepts to represent more complex or abstract ideas

Metaphor in Design

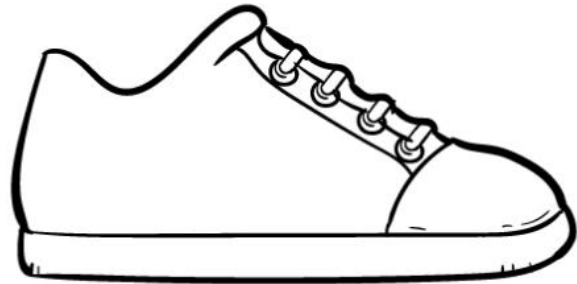
VS

Analogy in Design

	Metaphor	Analogy
Primary Purpose	Communication	Ideas Generation
Relationship	Direct Equivalence	Shared Principles
Focus	Understanding	Innovation

How Analogy Works in Design

Innovate the shoe

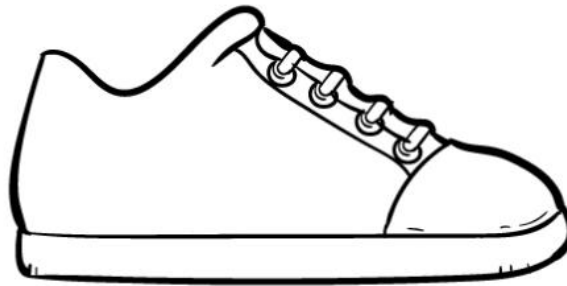


How Analogy Works in Design

Step 1: List down all the pain points

Gets dirty

Uncomfortable

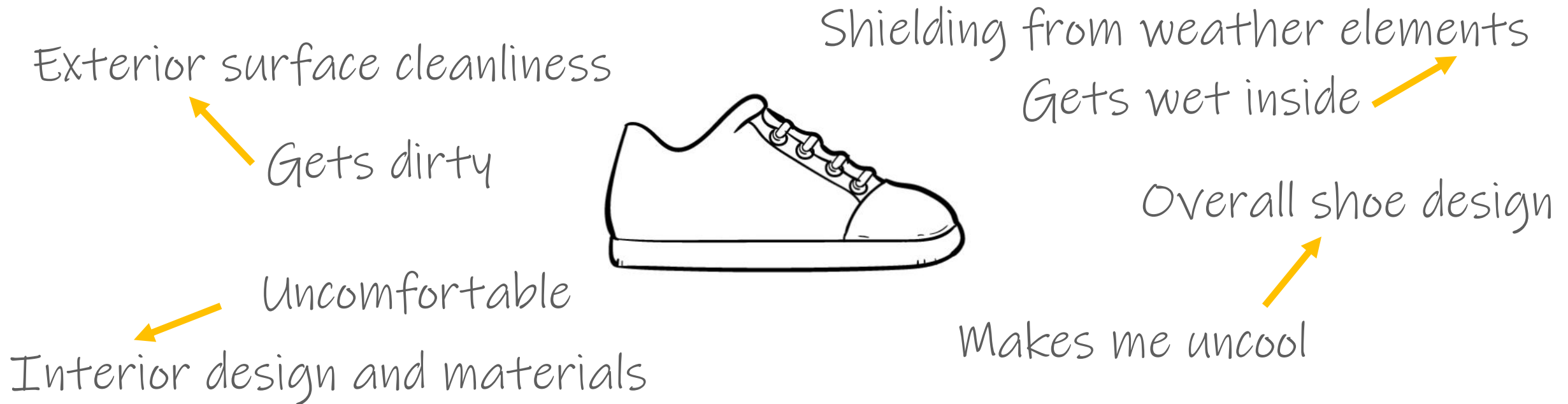


Gets wet inside

Makes me uncool

How Analogy Works in Design

Step 2: Convert the pain points to attributes



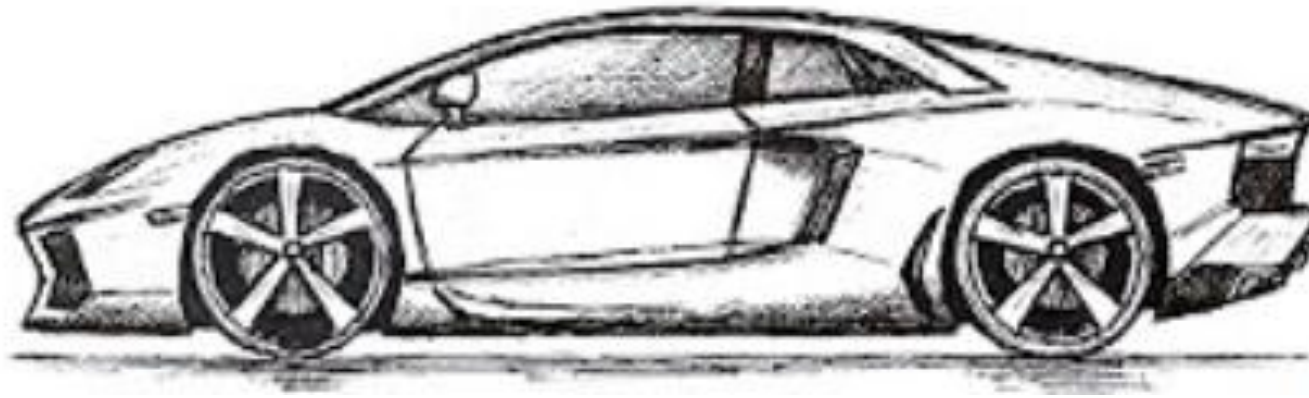
How Analogy Works in Design

Step 3: Identifying the Source

Selecting a source domain - a system, process, or phenomenon outside of your design problem. This could be anything from nature to technology, art to social systems.

How Analogy Works in Design

Step 3: Identify the Source



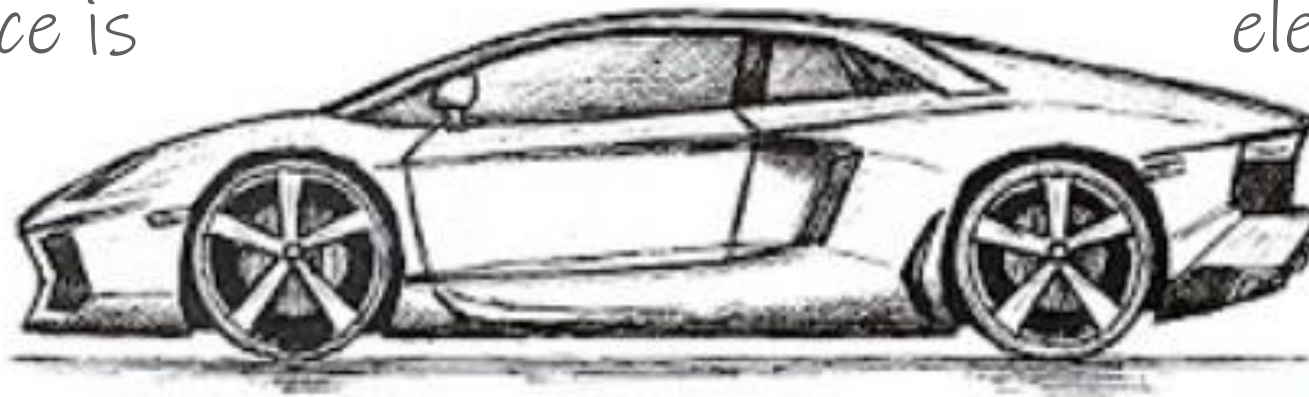
How Analogy Works in Design

Step 4: Mapping Relationships: Map the relationships and underlying principles of the source domain to the target domain - the design problem at hand. This involves identifying similarities in structure, function, or behaviour.

How Analogy Works in Design

Step 4: Mapping Relationships

Exterior surface is
dirt resistant



Interior design and materials
are comfortable

Interior is shielded
from weather
elements

Overall car design
is 'cool'

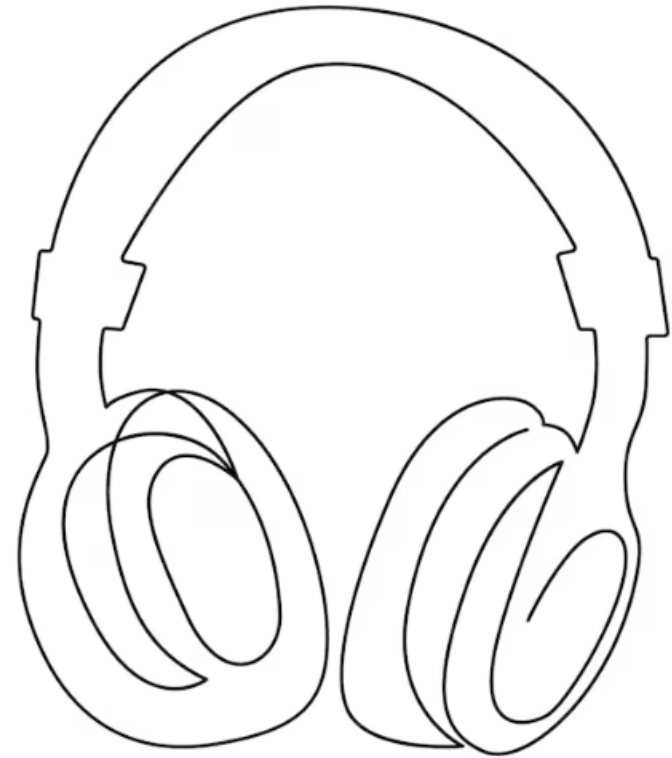
How Analogy Works in Design

1 Problems and Pains <i>Reflect again on the biggest problems and pains.</i> a. Gets dirty c. Makes me uncool b. Gets wet inside d. Uncomfortable	
Attributes <i>List possible attributes.</i> a. Exterior surface's cleanliness b. Shielding from weather elements c. Overall shoe design d. Interior design and materials	Analogies <i>Find analogies to the attributes.</i> a. Exterior surface is dirt resistant b. Interior is shielded from weather elements c. Overall car design is 'cool' d. Interior design and materials are comfortable

Note:
In this example, only 1 source is used.

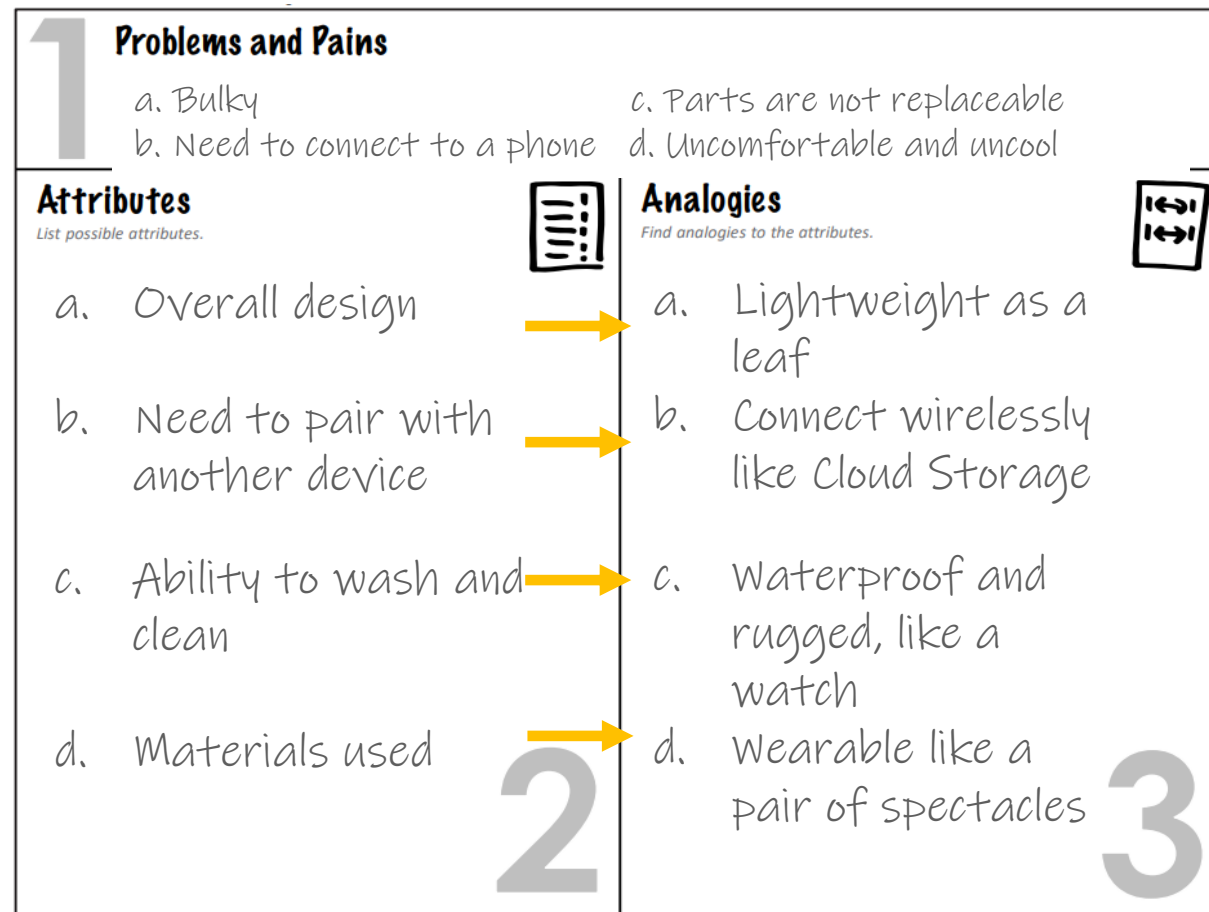
How Analogy Works in Design (Another example)

Innovate the headphone



How Analogy Works in Design (Another example)

Innovate the headphone



Reframing Opportunities:

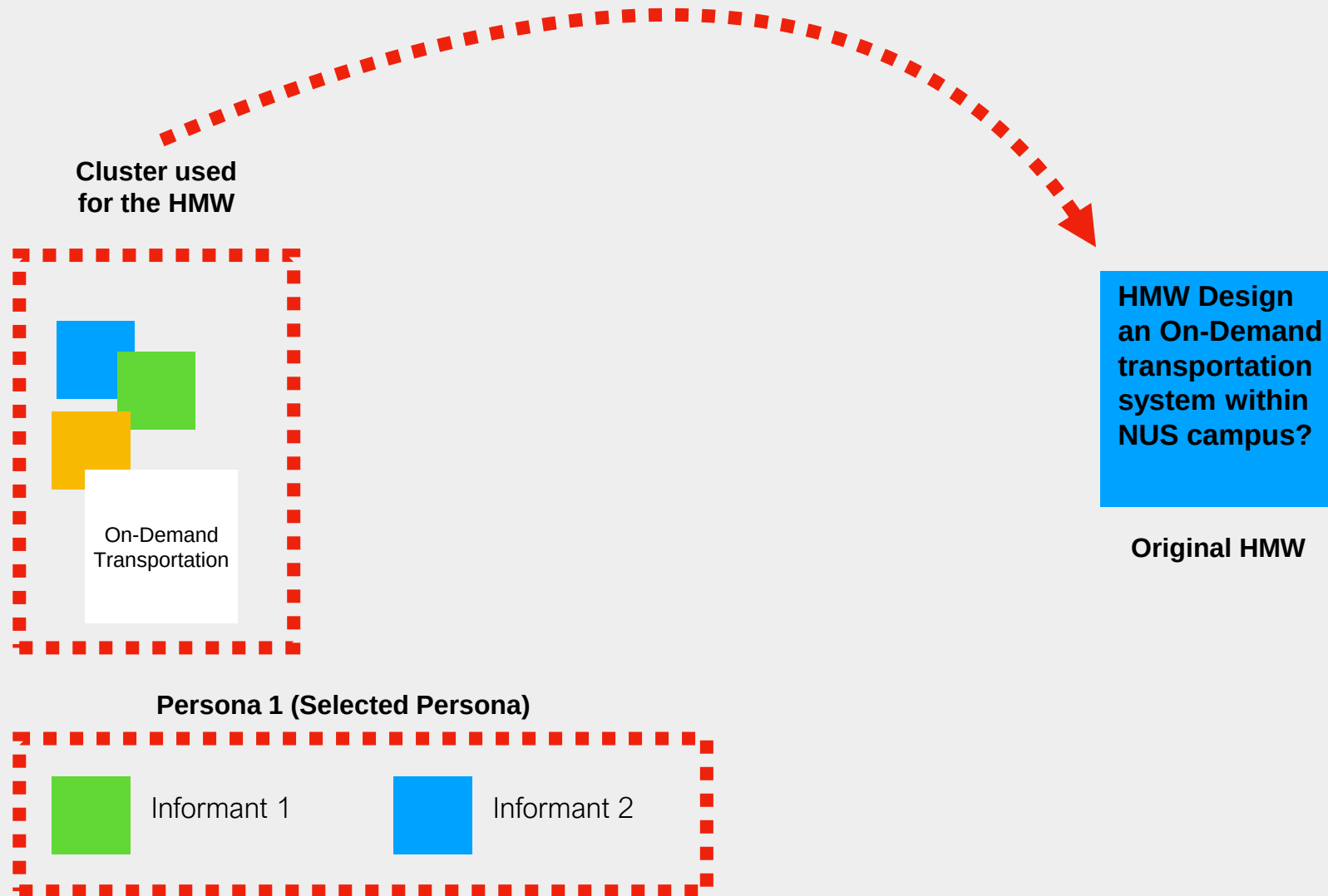
Powers of Ten

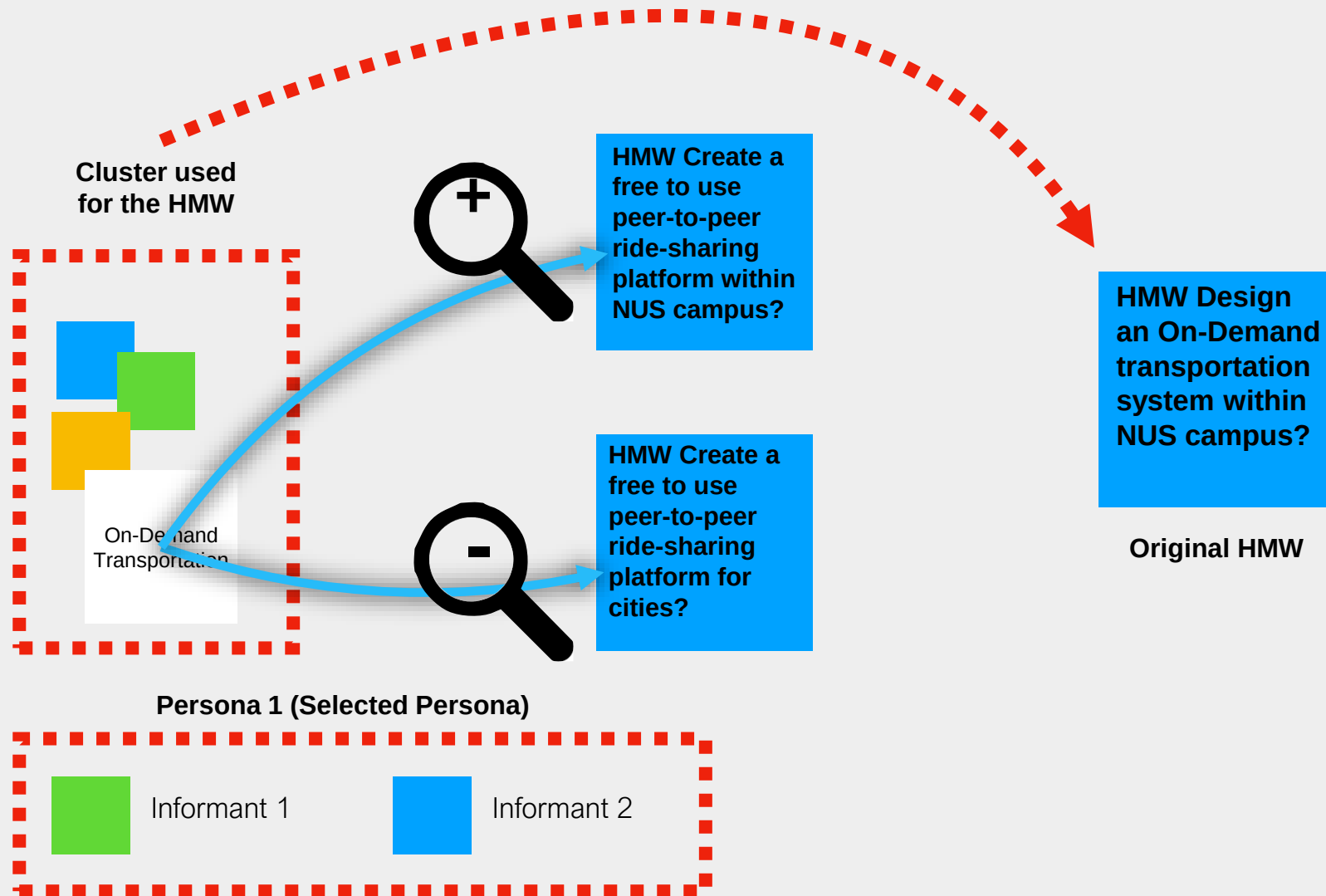
1. The Big Picture
2. Exploring Opportunities: **How Might We**
3. Reframing Opportunities: **Powers of Ten**
4. Problem Framing
5. Q&A

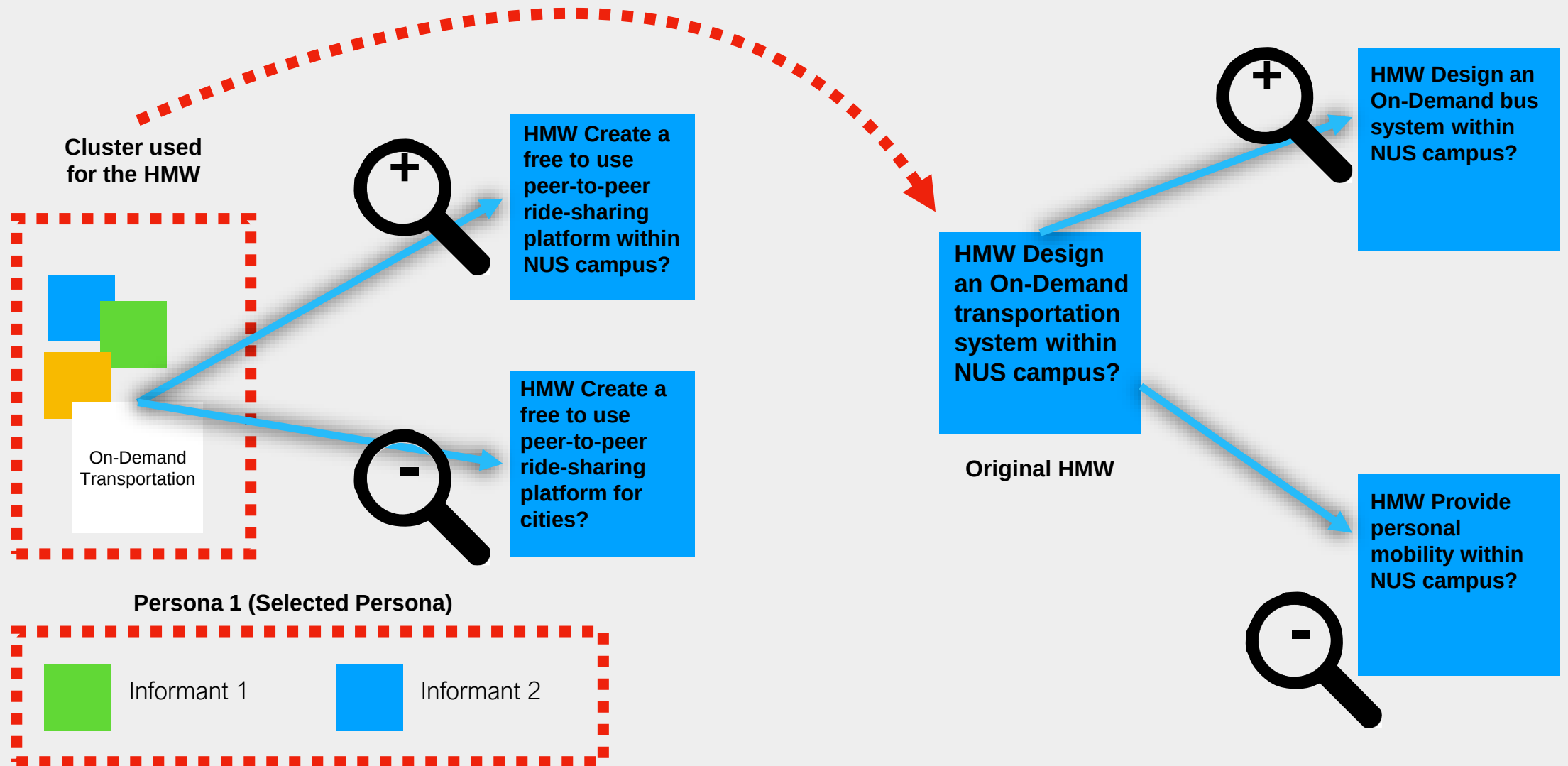
Powers of Ten is a reframing technique that can be used as a synthesis or ideation method. It allows the design team to use an intentional approach to **considering the problem at varied magnitudes** (*Increasing or Decreasing magnitudes*).

Powers of Ten helps you to look at every situation in the world from different angles, from close up, from far away, from upside down, and from behind.

Powers of Ten technique is inspired by the 1968 documentary film Powers of Ten, written and directed by Ray and Charles Eames.





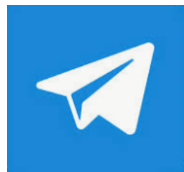


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KZ Tang